Monthly data, r=6, s=6

$$y_t = \alpha_0 + (\alpha_1 y_{t-1} + \sum_{j=1}^{5} \delta_j \Delta y_{t-j}) + \beta_1 z_t + \beta_2 p_t + \sum_{j=0}^{5} \gamma_{1j} \Delta z_{t-j} + \sum_{j=0}^{5} \gamma_{2j} \Delta p_{t-j}$$

	Estimate	Std. Error	t Value	Pr(>|t|)
α_0	0.172199	0.082234	2.094	0.036766
α_1	0.984146	0.012599	78.116	< 2e-16
δ_1	-0.561906	0.045781	-12.274	< 2e-16
δ_2	-0.232197	0.051549	-4.504	8.31e-06
δ_3	-0.197232	0.051835	-3.805	0.000160
δ_4	-0.183535	0.051397	-3.571	0.000390
δ_5	-0.084216	0.045048	-1.869	0.062151
β_1	0.024767	0.013131	1.886	0.059855
β_2	-0.007810	0.002865	-2.726	0.006637
γ_{10}	0.282465	0.095169	2.968	0.003142
γ_{11}	0.078767	0.097355	0.809	0.418860
γ_{12}	0.158477	0.098463	1.610	0.108141
γ_{13}	0.144970	0.098085	1.478	0.140044
γ_{14}	0.014257	0.097219	0.147	0.883470
γ_{15}	0.015178	0.094403	0.161	0.872335
γ_{20}	-0.175451	0.028035	-6.258	8.44e-10
γ_{21}	-0.029226	0.031245	-0.935	0.350045
γ_{22}	-0.037857	0.031734	-1.193	0.233455
γ_{23}	-0.003084	0.032193	-0.096	0.923709
γ_{24}	-0.060924	0.031788	-1.917	0.055868
γ_{25}	0.021312	0.029506	0.722	0.470445

income elasticity $\varepsilon_I = \dfrac{\beta_1}{1-\alpha_1} = \dfrac{0.024767}{1-0.984146} = 1.5622$

price elasticity $\varepsilon_P = \dfrac{\beta_2}{1-\alpha_1} = \dfrac{-0.007810}{1-0.984146} = -0.4926$

Quarterly data, r=2, s=2.

$$y_t = \alpha_0 + (\alpha_1 y_{t-1} + \delta_1 \Delta y_{t-1}) + \beta_1 z_t + \beta_2 p_t + \sum_{j=0}^{1} \gamma_{1j} \Delta z_{t-j} + \sum_{j=0}^{1} \gamma_{2j} \Delta p_{t-j} + u_t$$

	Estimate	Std. Error	t value	p-value
α_0	-0.3611	0.0864	-4.178	4.48E-05***
α_1	0.9348	0.0158	59.118	< 2e-16 ***
δ_1	-0.0745	0.0691	-1.079	0.282
β_1	0.1266	0.0292	4.333	2.38E-05***
β_2	-0.0346	0.0077	-4.472	1.33E-05***
γ_{10}	-0.1654	0.1230	-1.345	0.18
γ_{11}	0.0451	0.1201	0.375	0.708
γ_{20}	-1.0380	0.0328	-31.621	< 2e-16 ***
γ_{21}	-0.0620	0.0822	-0.754	0.452

income elasticity $\varepsilon_I^{'} = \dfrac{\beta_1}{1-\alpha_1} = \dfrac{0.126619}{1-0.934802} = 1.942069$

price elasticity $\varepsilon_P^{'} = \dfrac{\beta_2}{1-\alpha_1} = \dfrac{-0.034594}{1-0.934802} = -0.5305991$

➔ $\varepsilon_I < \varepsilon_I^{'},\ \varepsilon_P < \varepsilon_P^{'}$

	monthly data		quarterly data	
	ε_I	ε_P	ε_I	ε_P
r=0, s=0	0.9780347	-0.1467188	1.793472	-0.4449148
r=1, s=1	0.9730371	-0.1462163	1.928016	-0.5247408
r=2, s=2	1.160423	-0.2567503	1.942069	-0.5305991
r=3, s=3	1.250695	-0.3065278	1.970166	-0.5419624
r=4, s=4	1.296404	-0.3388186	1.988834	-0.5492182
r=5, s=5	1.470007	-0.4418323	1.965582	-0.5298847
r=6, s=6	1.562193	-0.4926202	2.024881	-0.5570184
r=7, s=7	1.588781	- 0.5055965	2.043544	- 0.5597218
r=8, s=8	1.541354	- 0.4953556	2.120272	- 0.5942209
r=9, s=9	1.498679	-0.4801856	2.115979	-0.6055438
r=10, s=10	1.589933	-0.5161333	2.064227	- 0.5752607

➔

(i) with quarterly data, always higher absolutely value of the elasticity than with monthly data.

(ii) when r and s is small, the larger r and s, the larger absolute value of the elasticity of income and price; however, when r and s become larger enough, this kind of trend is disappear – especially, with the quarterly data, the order is quite random.